THE UNIVERSITY OF WESTERN AUSTRALIA

[CITS5206] INFORMATION TECHNOLOGY CAPSTONE PROJECT

# PROJECT SPEC AND PLANS

Group 05


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CONTENTS


**I Problem Statement** 1
- I-A Problem Definition . . . 1
- I-B Current Solution and Limitations . . . 1
  - I-B1 Existing Solution . . . 1
  - I-B2 Limitations . . . 1
- I-C Clients' Needs & Justifications . . . 2
  - I-C1 Functional Requirements . . . 2
  - I-C2 Non-Functional Requirements . . . 2
- I-D Expected Deliverable . . . 2
- I-E Success Criteria . . . 3

**II Client Communication and MVP Agreement** 4
- II-A Stakeholder Identification . . . 4
- II-B Communication Methods . . . 4
- II-C Client Requirements Summary . . . 4
- II-D Minimum Viable Product (MVP) . . . 5
- II-E Feedback and Iteration Approach . . . 6

**III Project Planning & Management** 6
- III-A Project Management Overview . . . 6
- III-B Team Responsibilities . . . 6
- III-C Project Methodology . . . 7
- III-D Task Breakdown . . . 7
  - III-D1 Requirement Finalisation . . . 7
  - III-D2 System Design . . . 7
  - III-D3 Backend Development . . . 7
  - III-D4 Frontend Development . . . 7
  - III-D5 Testing & Deployment . . . 7
- III-E Timeline & Milestones . . . 7
- III-F Tools & Tracking . . . 8
  - III-F1 Version Control & Task Management . . . 8
  - III-F2 Communication . . . 8
  - III-F3 Documentation . . . 8
  - III-F4 Progress Tracking . . . 8

**IV Risk and Technology Assessments** 8
- IV-A Risk Assessment . . . 8
- IV-B Skills and Resources Analysis . . . 9
- IV-C Technology Choices and Justification . . . 10
  - IV-C1 Selected Solution . . . 10
  - IV-C2 Alternative Options Considered . . . 10

**V Appendix** 10
- V-A Client Interview Q&A . . . 10

# WMAA Charity Website Redevelopment Project

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**Abstract**

This document presents the project specification and planning for the WA Multicultural Assistance Alliance (WMAA) Charity Website Redevelopment Project. It outlines the current problems with the existing website, the proposed solution, system requirements, client communication process, agreed minimum viable product (MVP), project planning and management strategy, and risk and technology assessments. The aim is to deliver a secure, user-friendly, and maintainable website that better supports WMAA's services, volunteer engagement, fundraising, and multilingual accessibility.

**Index Terms**

Web application architecture, Flask, Frontend-backend integration, Responsive web design, API integration, Secure payment processing, Multilingual systems, Form validation

## I. Problem Statement

### A. Problem Definition

The WA Multicultural Assistance Alliance (WMAA) is a non-profit organisation supporting multicultural communities through crisis housing, domestic violence support, fundraising, and community outreach. However, the current website does not effectively represent the organisation's services or provide an efficient and engaging platform for users.

The existing system lacks modern functionality and is difficult for non-technical staff to manage. The website is not fully user-friendly and provides a suboptimal user experience, with issues such as poor navigation and overlapping header elements affecting usability. Additionally, there is no sign-up or login functionality, limiting user interaction and volunteer engagement. The platform also does not adequately support key features such as dynamic content updates, structured volunteer management, and secure donation processing. These limitations reduce the effectiveness of communication, user engagement, and overall operational efficiency.

### B. Current Solution and Limitations

#### 1) Existing Solution:

- A basic informational website presenting WMAA services
- Existing donation workflow embedded within the website
- Static content with limited flexibility for updates

#### 2) Limitations:

- Difficult for staff and volunteers to update content without technical knowledge
- Limited support for dynamic features such as news updates and event galleries
- Lack of advanced donation features (e.g., automated receipts and acknowledgements)
- No structured system for volunteer registration or user accounts
- Limited scalability for multilingual support and future enhancements
- User experience and accessibility require improvement

### C. Clients' Needs & Justifications

WMAA Team, requires a redesigned and dependable website that preserves its existing organisational identity while improving digital functionality, user engagement, and long-term maintainability. The project is intended to deliver a structured online platform that supports the organisation's key operational and communication needs, including updated branding and contact information, clearly organised website pages, donation processing, volunteer registration, contact and newsletter features, and multilingual accessibility. These expectations are reflected in the functional requirements, which define the specific capabilities the system must provide.

In addition, the WMAA Team expects the website to perform effectively, securely, and consistently across different devices while remaining easy to manage and adaptable for future growth. These expectations are captured in the non-functional requirements, which focus on qualities such as usability, maintainability, security, performance, responsiveness, reliability, scalability, cost efficiency, and auditability.

#### 1) Functional Requirements:

| ID | Requirement Description |
|---|---|
| FR1 | Maintain existing branding elements (logo, colours, contact details) |
| FR2 | Support domain transition to `wmaacharity.com.au.org` |
| FR3 | Update address details as specified by the client |
| FR4 | Provide structured website pages (Home, About, Services, News & Gallery, Contact/Donate) |
| FR5 | Implement secure donation system with automated acknowledgement and invoice |
| FR6 | Include Terms & Conditions agreement and separate page |
| FR7 | Maintain existing donation workflow |
| FR8 | Enable volunteer registration with login/signup |
| FR9 | Integrate contact forms with anti-spam protection |
| FR10 | Support newsletter subscription |
| FR11 | Display government accreditation logos |
| FR12 | Provide multilingual support |

#### 2) Non-Functional Requirements:

| ID | Category | Requirement Description |
|---|---|---|
| NFR1 | Usability | Easy navigation and accessible design |
| NFR2 | Maintainability | Simple CMS for non-technical users |
| NFR3 | Security | Secure payments and data protection |
| NFR4 | Performance | Fast loading and smooth operation |
| NFR5 | Responsiveness | Works across mobile, tablet, desktop |
| NFR6 | Reliability | Minimal downtime |
| NFR7 | Scalability | Supports future enhancements |
| NFR8 | Cost Efficiency | Low-cost solution for a non-profit |
| NFR9 | Auditability | Trackable changes and updates |

By fulfilling the above requirements, the proposed website solution will provide substantial organisational and operational benefits to the WMAA Team by addressing the limitations of the current website and improving the overall quality of organisation's online presence and provide a scalable foundation for future growth and continued community outreach.

### D. Expected Deliverable

This project aims to enhance the existing WMAA website by implementing a set of front-end and back-end improvements based on the client's requirements. The expected deliverable is a functional and user-friendly website that improves usability, supports essential organisational activities, and maintains consistency with WMAA's existing branding and content identity.

The final system is expected to include the following key deliverables:

- Migration and setup of the website under the new domain `wmaacharity.com.au.org`
- Update of address and other required organisational details
- Retention of existing branding elements, including logo, colour scheme, and contact information
- Clear website structure with core pages such as Home, About Us, Services/Programs, News & Gallery, and Contact/Donate
- Secure donation system with payment processing, automated confirmation email, and invoice generation
- Terms & Conditions agreement before payment, along with a dedicated Terms & Conditions page

In addition to these core website improvements, the project will also deliver several supporting features that improve engagement and accessibility. These features are intended to strengthen communication with users, simplify volunteer participation, and ensure the website is more inclusive for a wider audience.

- Volunteer registration system with signup and login functionality
- Contact enquiry form integrated with anti-spam protection
- Display of government and charity accreditation logos
- Initial multilingual support to improve accessibility for users from diverse language backgrounds

Overall, the expected deliverable is a more modern, functional, and accessible WMAA website that addresses the client's operational needs while providing an improved experience for donors, volunteers, and general users.

### *E. Success Criteria*

The success of the proposed website redevelopment will be evaluated against the following measurable criteria, aligned with the client's functional and non-functional requirements.

**Functional Success Criteria**

- All core website pages, including Home, About Us, Services/Programs, News & Gallery, and Contact/Donate, are implemented and accessible.
- The donation system processes test transactions securely and sends an acknowledgement email and invoice after each successful donation.
- A Terms and Conditions agreement is shown before payment, with a separate Terms and Conditions page available on the website.
- Users are able to complete volunteer registration, signup, and login successfully.
- The contact form accepts valid submissions.
- Government charity accreditation logos are displayed in the required locations.
- The website supports English and at least one additional language.

**Non-Functional Success Criteria**

- The website works correctly across desktop, tablet, and mobile devices without layout issues.
- Navigation is clear and allows users to access key pages and functions easily.
- Sensitive user information is handled securely, and payment processing is carried out through a secure third-party provider.
- The system is maintainable for non-technical staff with minimal technical assistance.
- The solution remains cost-effective and practical for long-term use by a non-profit organisation.
- The website performs reliably under normal usage with minimal downtime.

By meeting these criteria, the proposed solution will provide WMAA with a secure, usable, and sustainable website that supports communication, fundraising, and volunteer engagement.

## II. Client Communication and MVP Agreement

### A. Stakeholder Identification

The key stakeholders involved in this project include:

- **Client:** WA Multicultural Assistance Alliance (WMAA).
- **Primary Users:** Community members, donors, volunteers, and partners.
- **Administrative Users:** WMAA staff and volunteers responsible for managing website content.
- **Development Team:** Student team responsible for designing and implementing the solution.
- **Facilitator:** Unit facilitator providing guidance and feedback throughout the project.

### B. Communication Methods

Effective communication has been maintained with the client, facilitator, and team members to ensure alignment with project goals and requirements.

- **Client Meetings:** Conducted to gather requirements, clarify expectations, and receive feedback on progress.
- **Microsoft Teams:** Used for ongoing team discussions, document collaboration, meeting minutes, and communication with the facilitator.
- **Microsoft Outlook (UWA Email):** Used for formal communication with the client, including providing updates and confirming decisions. It is documented after each meeting to capture key discussions, agreed decisions, and action items.
- **GitHub:** Used for version control, issue tracking, and managing project tasks and progress. The repository also stores client meeting records and facilitator meeting records for project documentation and traceability [1], [2].

Regular communication through these channels ensures that all stakeholders remain informed and that project development stays aligned with client expectations.

### C. Client Requirements Summary

Based on discussions with the client, the following requirements were identified, covering both system functionality and performance expectations:

The website requires several front-end and back-end improvements to enhance usability, navigation, and overall user experience, addressing current issues such as poor layout, overlapping elements, and limited interactivity. The system must support a domain update from `wmaacharity.com.au` to `wmaacharity.com.au.org` and update the organisation's address while ensuring that existing branding elements, including logo, colours, and contact details, remain unchanged.

A key priority for the client is improving the online fundraising functionality. The donation feature must remain on the same page but be enhanced to provide a smoother and more secure user experience. This includes implementing a secure payment system that protects user data and ensures reliable transaction processing. After each successful donation, the system should automatically send an email acknowledgement and invoice to the donor. Additionally, a Terms and Conditions agreement popup must be displayed before payment, along with a separate Terms and Conditions page for transparency.

The website should also include a volunteer registration system with login and signup functionality to support better user engagement and management. Furthermore, government charity accreditation logos must be displayed prominently, particularly at the top of the website and near the donation section, to build trust and credibility.

To support WMAA's diverse community, the website should provide multilingual functionality, allowing users to access content in English and at least one additional language. The system should be user-friendly, responsive, and accessible across different devices, ensuring a consistent experience on mobile, tablet, and desktop platforms.

From a quality perspective, the system must be secure, reliable, and easy to maintain, allowing non-technical staff to manage content without difficulty. It should also be cost-effective, scalable for future enhancements, and capable of tracking key activities such as donations and updates for accountability.

TABLE I: Communication Plan

| Week | Task | Method | Purpose | Participants |
|---|---|---|---|---|
| 2 | Team formation and project selection | Microsoft Teams | Align team roles and select project | Team members |
| 3 | Initial client meeting request | Email (UWA Outlook) | Establish client contact and schedule meeting | Team, Client (Xiao Liang) |
| 4 | Client meeting – requirements gathering | Online meeting (Teams) | Gather detailed requirements and clarify scope | Team, Client |
| 5 | D1: Project Specification and Plan Submission | LMS | Submit finalised documents and confirm alignment | Team |
| 6–7 | MVP feature development and client demo | GitHub, Teams, Live Demo | Begin core implementation; present MVP and gather feedback | Team, Client |
| 8 | Update MVP based on client feedback | GitHub, Teams messages | Apply revisions per client input | Team |
| 9 | D2: Software Features Submission | LMS | Demonstrate feature completion, collect approval | Team, Client |

TABLE II: Client Requirements Traceability Table

| Category | Details | Source |
|---|---|---|
| **Current Process** | Staff manually update a static website; no CMS or dynamic content management exists. | Client meeting, Week 4 |
| **Main Users** | WMAA staff, volunteers, donors, and community members from diverse cultural backgrounds. | Client meeting, Week 4 |
| **Donation System** | Secure payment with automated email acknowledgement and invoice. Terms & Conditions popup before payment. | Client: "We need receipts sent automatically after each donation." |
| **Volunteer Management** | Signup and login system for volunteers to register and manage their participation. | Client meeting, Week 4 |
| **Multilingual Support** | Content available in English and at least one additional language. WMAA serves diverse cultural communities. | Client: "Our community speaks many languages." |
| **Accreditation Logos** | Government charity accreditation logos to be displayed prominently near the donation section. | Client meeting, Week 4 |
| **Domain & Branding** | Transition to `wmaacharity.com.au.org`; retain existing logo, colours, and contact details. | Client email confirmation |

### *D. Minimum Viable Product (MVP)*

The Minimum Viable Product (MVP) defines the essential features that will be delivered to meet the client's core requirements.

The agreed MVP includes:

- Updated website with improved user interface and navigation
- Domain update and basic content updates (address, structure)
- Functional donation system with secure payment and confirmation email
- Implementation of Terms & Conditions agreement and separate page
- Basic volunteer registration system (signup/login)
- Improved form security using HTML, CSS, and JavaScript, including proper input checks and preventing incorrect or unwanted data submission.
- Display of accreditation logos
- Initial support for multilingual functionality

These features ensure that the website addresses the most critical issues while remaining feasible within the project timeline.

### *E. Feedback and Iteration Approach*

The project follows an iterative approach where feedback from the client and facilitator is continuously incorporated.

- Feedback is collected after meetings and presentations
- Changes are prioritised based on client needs and project scope
- Minor updates are implemented during development cycles
- Major changes are discussed and evaluated before implementation

This approach ensures that the final system remains aligned with client expectations while maintaining project feasibility.

## III. Project Planning & Management

### *A. Project Management Overview*

The objective of this project is to design and develop an enhanced web-based platform for the WA Multicultural Assistance Alliance (WMAA), addressing current limitations in usability, functionality, and user engagement identified in Section 1. The system will incorporate key MVP features such as secure donation processing, volunteer registration, multilingual support, and improved navigation.

To ensure successful delivery, the project is structured into iterative development phases aligned with deliverable timelines. The immediate focus for Deliverable 2 is to transition from planning to implementation by developing a functional prototype of the system.

The key goals for the next development phase include:

- Finalizing system architecture
- Developing core backend features such as authentication and donation workflows
- Implementing responsive frontend pages aligned with the approved UI structure
- Deploying an initial working version of the application for client feedback

This structured and incremental approach ensures that development remains aligned with client expectations while allowing flexibility for refinement.

### *B. Team Responsibilities*

Responsibilities are assigned based on team members' demonstrated contributions and strengths observed during Weeks 2–4, ensuring that each member has clear ownership of tasks for the next phase.

TABLE III: Team Responsibilities for the Next Phase

| **Role** | **Student Name** | **Responsibilities (Next Phase)** |
|---|---|---|
| Project Coordinator | Aneesh Kumar | Manage client communication, schedule meetings, track progress, ensure milestone completion |
| Frontend Lead | Navdeep Singh | Design UI structure, implement responsive pages, ensure usability and accessibility |
| Documentation & QA | Varun Suresh | Maintain documentation, track requirements, record meeting minutes, perform testing |
| Backend Developer | Harpreet Singh | Develop backend logic, authentication system, API integration |
| Integration & Review | Neha Pathare | Support backend/frontend integration, review code, ensure quality and consistency |

These responsibilities are derived from actual team contributions such as communication coordination, repository setup, documentation drafting, and quality review, ensuring realistic and evidence-based planning.

### C. Project Methodology

An Agile (iterative and incremental) methodology is adopted due to evolving requirements and initial uncertainty in client communication.

During Weeks 3–4, the team experienced delays in meeting the client, which required continued progress despite incomplete requirements. This reinforced the need for a flexible methodology that allows adaptation.

### D. Task Breakdown

To meet Deliverable 2 requirements, tasks are broken down into actionable components:

*1) Requirement Finalisation:*

- Consolidate all client requirements from meetings
- Define detailed functional specifications (donation, login, multilingual support)
- Create user flow diagrams for system navigation

*2) System Design:*

- Define frontend-backend architecture
- Design database schema (users, donations, volunteers)
- Select APIs for payment processing (e.g., Stripe or similar)

*3) Backend Development:*

- Implement authentication system (login/signup)
- Develop donation processing logic with secure transactions
- Create APIs for data handling and integration

*4) Frontend Development:*

- Develop responsive UI pages (Home, About, Services, Donate, Contact)
- Implement form validation and user interaction features
- Integrate frontend with backend APIs

*5) Testing & Deployment:*

- Conduct unit and integration testing
- Deploy prototype (e.g., Vercel or similar platform)
- Share system with client for feedback

This task breakdown ensures that each team member has clear, actionable tasks, directly satisfying the rubric requirement for detailed planning.

### E. Timeline & Milestones

The project timeline follows a structured progression aligned with unit deliverables.

| **Time Period** | **Milestone / Activity** |
|---|---|
| Weeks 1–2 | Team formation, project selection, initial planning |
| Weeks 3–4 | Requirement gathering, delayed client meeting, Deliverable 1 preparation |
| Week 5 | Deliverable 1 submission |
| Weeks 6–7 | Backend and frontend development (core features) |
| Week 8 | Prototype deployment and internal testing |
| Week 9 | Deliverable 2 submission (functional system demo) |
| Weeks 10–11 | Refinement based on feedback |
| Week 12 | Final system delivery and presentation |

The timeline ensures a realistic progression from planning to implementation, with buffer time for testing and revisions.

### F. Tools & Tracking

The team uses a structured set of tools to manage development and collaboration.

*1) Version Control & Task Management:*

- **GitHub Repository**
  - Tracks code changes and version history
  - Issues used to define tasks and assign responsibilities
  - Project board used with columns: *To Do* → *In Progress* → *Review* → *Done*
  - The project source code, version history, and task tracking are maintained in the team GitHub repository [3].

*2) Communication:*

- **Microsoft Teams**
  - Weekly meetings and quick discussions
  - File sharing and collaboration
- **Email (Outlook)**
  - Formal communication with client and facilitator

*3) Documentation:*

- **OneDrive / Shared Docs**
  - Store deliverables, meeting minutes, and requirements

*4) Progress Tracking:*

- **Weekly Accountability Reports**
  - Track individual contributions and ensure transparency
  - Provide evidence of consistent team engagement

## IV. Risk and Technology Assessments

The team conducted a structured assessment of project risks, team skills, and technology options before beginning development. Each risk is paired with a concrete mitigation strategy, and all technology decisions are justified against the project's specific context: a full website rebuild for a non-profit organisation using Flask on the backend, HTML/CSS/JS on the frontend, and Stripe for payment processing as confirmed by the client.

### A. Risk Assessment

| **Risk** | **Description and Mitigating Measures** |
|---|---|
| **No Existing Codebase to Work From** | The team initially assumed the existing WMAA website could be shared for reference. The client confirmed via email that the site runs entirely on Wix, meaning there is no source code or database available. The project is therefore a full rebuild rather than an enhancement.<br>*Mitigating measures:* The project is treated as a greenfield build from the outset. Development stays focused strictly on the agreed MVP features. Fortnightly client meetings are used to continuously validate that what is being built matches expectations before too much development occurs in the wrong direction. |

| Risk | Description and Mitigating Measures |
|---|---|
| **Scope Creep from Additional Client Requests** | The client has expressed interest in features beyond the core MVP, such as support for all languages and newsletter functionality. Without clear boundaries, these additions could delay delivery of the core system.<br>*Mitigating measures:* The agreed MVP serves as the fixed baseline. Any new requests raised during meetings are logged and assessed before development begins. Features outside the agreed scope are either scheduled for a later phase or formally excluded with client acknowledgement. |
| **Security of User Data** | The system collects and processes sensitive information including donor payment details, volunteer personal data, and contact form submissions. A security failure could harm users and damage the organisation's reputation.<br>*Mitigating measures:* All security logic is handled exclusively in the Flask backend. This includes input validation and sanitisation on every form, CSRF protection using Flask-WTF, session management with secure HttpOnly cookies, and password hashing using Werkzeug. The HTML/CSS/JS frontend is limited to presentation only, meaning no sensitive data is processed or stored client-side. Since Stripe Checkout is used, card data never passes through the application itself. |
| **Payment Integration Complexity** | Stripe has been confirmed by the client as their existing payment provider. Correctly handling Stripe webhooks, which are required to trigger automated donation confirmation emails and invoice generation, carries implementation risk if not planned carefully.<br>*Mitigating measures:* The team will use Stripe Checkout, where the payment page is hosted entirely on Stripe's own servers. This removes the most technically demanding part of payment handling from the team's codebase. All webhook events received by Flask will be verified using Stripe's signature validation before any action is triggered. Payment flows will be tested thoroughly in Stripe's sandbox environment before any live deployment. |
| **Timeline Delays** | The team encountered a delayed client meeting during Weeks 3–4, which led to an approved extension for Deliverable 1. Similar delays during the development phase could place the Deliverable 2 deadline at risk.<br>*Mitigating measures:* The team uses fortnightly sprints with clearly defined goals tied to each deliverable. A GitHub project board is used to track task status across all members so blockers can be identified and addressed early. Buffer time is built into the schedule ahead of each submission deadline. |

### *B. Skills and Resources Analysis*

A skills review was conducted before responsibilities were assigned. The team has solid coverage across the areas required: frontend development using HTML, CSS, and JavaScript, backend development using Flask, and documentation, testing, and QA.

Areas requiring additional preparation were identified as: Flask security configuration (Flask-WTF, Werkzeug, Flask-Talisman), Stripe webhook integration within Flask, and multilingual content switching implementation. The client specifically referenced the Australian Government website as a model for multilingual design, which gives the team a concrete benchmark to work towards.

To address these gaps, relevant team members have been assigned to review Flask's official security documentation and Stripe's development documentation. Team members can commit approximately 20–25 hours per week to the project and have access to the hardware, software, and university resources required.

### C. Technology Choices and Justification

*1) Selected Solution:* The team selected a practical stack that matches existing skills, meets the client's requirements, and produces a system that is simple to maintain after handover — important for a non-profit with limited technical staff.

- **HTML, CSS, JavaScript (Frontend):** Standard web technologies are used without a heavy JavaScript framework. This keeps the codebase lightweight and easy to hand over post-project. JavaScript handles only interactivity — form behaviour, the multilingual toggle, and the Terms & Conditions popup. No sensitive logic runs client-side.
- **Flask (Backend):** Flask handles all routing, form processing, user authentication, session management, Stripe webhook events, and email dispatch. Its simplicity keeps the codebase auditable and maintainable. All application logic sitting server-side in Flask means security is managed in one centralised layer.
- **Flask Security Layer:** All security measures are implemented in Flask rather than on the frontend. This covers input validation and sanitisation on every form, CSRF protection via Flask-WTF, secure session management with HttpOnly cookies, password hashing via Werkzeug, and HTTPS enforcement in production via Flask-Talisman. CAPTCHA-based anti-spam protection will be applied to all public-facing forms including contact and volunteer registration.
- **Stripe Checkout (Payment Processing):** The client confirmed Stripe as their existing payment provider. Stripe Checkout redirects users to a Stripe-hosted payment page, meaning card data never touches the team's frontend or backend. Once payment is completed, Stripe sends a webhook POST to a Flask endpoint. Flask verifies the webhook signature using the Stripe-provided secret before triggering the automated confirmation email and invoice to the donor. The Stripe secret key is stored as an environment variable and never hardcoded.
- **Hosting:** A low-cost platform supporting Flask deployment will be used. The platform will enforce HTTPS, support environment variable management, and provide sufficient reliability at minimal cost for a non-profit.

*2) Alternative Options Considered:*

- **WordPress with plugins:** Familiar to non-technical users but plugin dependencies introduce ongoing security risks and remove direct control over backend security — a core requirement for this project.
- **Django (Backend):** More feature-complete than Flask but significantly heavier and more complex than the project requires. Flask gives the team direct control over exactly what is built and how it is secured, without unnecessary overhead.
- **Wix (existing platform):** The client's current site cannot support custom backend code, making Flask-based security, Stripe webhook handling, and volunteer authentication all impossible. A custom rebuild is the only viable path.

## V. Appendix

### A. Client Interview Q&A

TABLE V: Client Interview Q&A

| ID | Question / Answer |
|---|---|
| Q1 | What is the client's expectation for the website redevelopment? |
| A1 | The client expects a brand-new website design rather than simply replicating the existing one. They described the current website as outdated and would prefer a more modern, attractive, and engaging design while still maintaining the organisation's identity. |
| Q2 | Will the website domain name remain the same? |

| ID | Question / Answer |
|---|---|
| A2 | No. The client indicated that the website address is expected to change from the current domain to `wmaacharity.com.au.org`. |
| Q3 | Which existing elements must be retained in the new website? |
| A3 | The client wants the current contact details, address, phone number, logo, and logo colours to remain unchanged. While the placement and size of the logo may be adjusted, its colours must stay the same because they represent the organisation's branding and trademark identity. |
| Q4 | What are the client's requirements for the donation functionality? |
| A4 | The donation function must remain available on the new website. The client wants users to continue being able to make donations online, and the existing payment methods should be maintained. |
| Q5 | What additional features are required for the donation system? |
| A5 | The client requested that after a successful online donation, the system should automatically send an acknowledgement email to the donor with an invoice attached. The client also emphasised that the donation page must be secure. |
| Q6 | What are the client's expectations regarding Terms and Conditions? |
| A6 | The client wants a standard Terms and Conditions page to be included on the website, similar to other organisational websites. It does not need to appear automatically when users first enter the website, but it should be available as a separate page or section for access when needed. |
| Q7 | Are there any accreditation or trust elements that must be displayed? |
| A7 | Yes. The client stated that the government charity accreditation logo is very important. It should be displayed on the home page as well as on the donation page to demonstrate that the organisation is officially registered and authorised. |
| Q8 | Does the client require multilingual support? |
| A8 | Yes. The client wants the website to support multiple languages, not only Chinese. They indicated that the language options should be similar to those available on Australian government websites and should cater to multicultural communities. |
| Q9 | What volunteer-related functionality does the client want? |
| A9 | The client wants a *Join Us* or volunteer sign-up function where interested users can provide their full name, contact number, and email address. This information should be submitted to the organisation so that they can contact volunteers when needed for future events or charity activities. |
| Q10 | Does the client want general user sign-up or membership functionality? |
| A10 | No formal membership functionality is required. The client clarified that the sign-up process is mainly intended for volunteers, not for a paid or business-style membership model. It is simply a way for people to register their interest in helping the charity. |
| Q11 | Should volunteer information be accessible to everyone? |

| ID | Question / Answer |
|---|---|
| A11 | No. The client confirmed that volunteer details submitted through the website should only be accessible to authorised administrators. |
| Q12 | Does the client want notifications for volunteer registrations? |
| A12 | Yes. The client indicated that when a new volunteer signs up, an automatic email notification should be sent to the organisation so they are informed of the registration and can record the volunteer's details in their database. |
| Q13 | Does the client require a separate help-request form on the website? |
| A13 | This was discussed during the meeting, but the client preferred to keep the website simple. They noted that the existing Contact Us section is already sufficient, so a separate help-request button is not essential. |
| Q14 | Will the client provide website content and media assets? |
| A14 | Yes. The client stated that they will provide additional photos, stories, and information about previous and ongoing projects for use in the new website. |
| Q15 | Can the team access the existing website resources or source code? |
| A15 | Yes. The client said they would provide contact details for the volunteer who originally built the current website, so that technical information and existing resources can be accessed if required. |
| Q16 | Is another student group working on the same project? |
| A16 | Yes. The client confirmed that another group is working on the same website redevelopment project, and both groups will receive the same requirements, updates, photos, and stories. |

## References


[1] Group 05. "Wmaa website project – client meetings." GitHub repository folder, accessed 29 March 2026. [Online]. Available: https://github.com/Harpreet-Personal/WMAA-Website-Project/tree/main/client-meetings.

[2] Group 05. "Wmaa website project – facilitator meetings." GitHub repository folder, accessed 29 March 2026. [Online]. Available: https://github.com/Harpreet-Personal/WMAA-Website-Project/tree/main/facilitator-meetings.

[3] WMAA Website Project Team, *Wmaa website project*, https://github.com/Harpreet-Personal/WMAA-Website-Project, GitHub repository, accessed 29 March 2026, 2026.